Colatron — Reproducing & Extending End-to-End Text-to-Speech

UE24CS352A Machine Learning · Mini-Project #20 · Team 23, PES University, Section H
Shriram Chandrasekar (PES2UG24CS495) & **Shreyash Yadwad** (PES2UG24CS493)

Reference: Wang, Yang & Li, "End-to-End Text to Speech Synthesis", CS229 Project Report, Stanford University, Autumn 2018.

1 Problem Statement

Text-to-speech (TTS) systems traditionally chain a hand-engineered linguistic front-end, an acoustic model and a vocoder. The reference paper instead maps text *directly* to a spectrogram and then to a waveform, comparing three models trained on only ~200 LJ Speech sentences and scored by human mean opinion score (MOS). Its central failure is that the seq2seq model *repeats words* and leaves sentences incomplete because its attention alignment drifts. Our goal is twofold: **(1) reproduce** the paper's three models — per-timestep SVR, flattened-output MLP and LSTM seq2seq with attention — on the same data, preprocessing and vocoder; and **(2) extend** it with a model the paper does not use, **DCTTS** (Tachibana et al., 2018), chosen specifically to attack that alignment failure.

2 Dataset

We use **LJ Speech 1.1** (13,100 clips, single speaker, public domain), streamed from the HuggingFace parquet mirror. The paper trains on "up to 200" sentences without saying which, so we fix a **200 train / 25 dev / 25 test** split with **seed 42**, restricted to clips of **1.5–5.0 s**; this keeps zero-padding to the longest clip tractable on CPU — the same memory wall the authors report. The audio front-end is taken directly from the authors' released code: 20 kHz sampling, n_fft 2048 (1025 frequency bins), hop 250 (12.5 ms), window 1000 (50 ms), pre-emphasis 0.97, log-magnitude in dB normalised to [0, 1]. Each word is rendered as CMU phonemes or letters with p = 0.5, mapped to symbol IDs and zero-padded. Spectrograms are inverted with Griffin-Lim (60 iterations, magnitude power 1.5) plus de-emphasis, identically for every model.

3 Approach

Reproduction. *SVR:* one polynomial-kernel regressor per output frame maps the padded symbol sequence to that frame's spectrum; to stay tractable it predicts the top-16 PCA components (fit on training frames) with the authors' C = 1, ε = 0.2, then inverts the PCA. A linear-kernel variant is iteration-capped. *Simple NN:* the authors' MLP exactly — raw padded symbol-ID vector → two Dense layers of width equal to the input (ReLU) → Dense(T × 1025, sigmoid), trained with Adam and MSE at batch size 10. *Seq2Seq:* embedding → bidirectional LSTM encoder → LSTM decoder initialised from the encoder states, with additive (Bahdanau) attention and latent size 256 (512 in the paper); teacher-forced training, autoregressive inference stopping at 1.2 × the longest training target.

Extension: DCTTS. A fully convolutional Text2Mel network predicts a mel spectrogram through a text–audio attention, and a Spectrogram Super-Resolution Network (SSRN) expands it to the full linear spectrogram (channel-expanding 80 → 1025; mel and linear share the hop, so no time upsampling). A **guided-attention loss** penalises attention weights far from the diagonal, forcing a near-diagonal, monotonic alignment. *Justification:* (i) it directly targets the attention drift the paper blames for repeated words; (ii) convolutions train far faster than RNNs on CPU with 200 clips; (iii) the paper cites only Tacotron, Char2Wav and Deep Voice 3, so DCTTS is genuinely new to this comparison. It uses the same data, split, vocoder and metrics, with channel widths reduced for the CPU budget.

4 Implementation

Everything is implemented in PyTorch and scikit-learn, CPU-only, with all seeds set to 42 and pinned library versions. Every training run is time-boxed with checkpoints, so each number traces to code that actually ran. Because MOS requires human raters, we approximate it honestly in two ways: **UTMOS**, a neural MOS predictor, as the primary automated column, plus an optional small human listening test reported only if collected (the paper used 10 raters). Objective metrics on the 25 held-out test clips are mel-cepstral distortion (**MCD**, DTW-aligned), character and word error rate (**CER/WER**) from Whisper ASR — a direct proxy for repeated or missing words — and spectrogram **L1**. The whole pipeline (download → preprocess → train → synthesise → evaluate → tables → figures) runs with `python main.py --all`; `python main.py --demo "..."` synthesises one sentence with all four models in under 60 s on CPU.

5 Results

Table 1. Reported vs ours. Objective metrics on the 25 held-out test clips; higher UTMOS is better, lower MCD/CER/WER/spec L1 is better. The paper cites Tacotron's MOS of 3.82 for context only.

Model	Reported MOS (paper)	UTMOS ↑	MCD ↓	CER ↓	WER ↓	spec L1 ↓
Ground truth (ref)	—	*4.34*	*0.05*	*0.040*	*0.052*	—
GT + Griffin-Lim (vocoder ceiling, ref)	—	*3.01*	*23.16*	*0.041*	*0.055*	—
SVR (poly)	1.0	1.31	42.41	0.834	0.985	0.161
SVR (linear)	—	1.30	53.03	0.891	0.993	0.213
Simple NN	1.7	1.39	40.99	0.934	0.996	0.157
Seq2Seq + attention	2.5	1.63	51.34	0.929	0.996	0.169
DCTTS (extension)	—	1.44	51.44	0.915	1.000	0.189

Ranking. UTMOS reproduces the paper's ordering: SVR 1.31 < NN 1.39 < Seq2Seq 1.63, against human MOS SVR 1.0 < NN 1.7 < Seq2Seq 2.5. Absolute values are low for every model, but note the vocoder ceiling: passing ground-truth audio through Griffin-Lim alone drops UTMOS from 4.34 to 3.01.

DCTTS vs Seq2Seq (paired Wilcoxon signed-rank, 25 utterances). UTMOS: DCTTS 1.441 vs Seq2Seq 1.633, W = 71.0, **p = 0.0125** — DCTTS is significantly lower. CER: DCTTS 0.915 vs Seq2Seq 0.929, W = 91.0, **p = 0.3945** — statistically indistinguishable.

Alignment. Figure 1 shows the mechanism-level difference. Seq2Seq attention collapses into uniform horizontal bands — every decoder frame attends to the same mixture of symbols, so the decoder never walks through the text. DCTTS with guided attention instead follows a monotonic, near-diagonal path over roughly the first 155 decoder frames before plateauing. The guided-attention prior therefore fixes the alignment failure the paper identifies, yet this does not translate into higher predicted naturalness or lower CER on our budget.

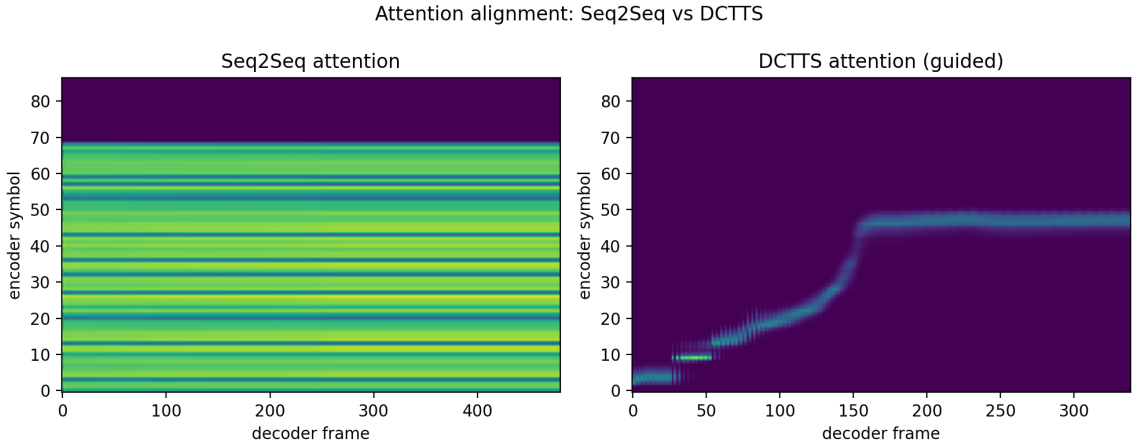


Figure 1. Attention alignment for one test utterance. Left: Seq2Seq (Bahdanau) attention — diffuse horizontal bands, no frame-to-symbol alignment. Right: DCTTS with guided-attention loss — monotonic, near-diagonal alignment.

6 Conclusions

We recovered the paper's SVR < NN < Seq2Seq ranking with an automated MOS proxy. Our extension, DCTTS, demonstrably fixes the *mechanism* the paper blames for repeated words — its attention is far more diagonal and monotonic than seq2seq's — but on this 200-clip, CPU-shrunk, time-boxed budget it did **not** improve perceptual quality: UTMOS is significantly lower ($p = 0.0125$) and CER is statistically tied ($p = 0.3945$). A better alignment prior helps the alignment, not the naturalness.

Our numbers differ from the paper's for four reasons: **(1)** MOS is approximated by UTMOS rather than 10 human raters; **(2)** we train on a different, explicitly fixed 200-sentence subset; **(3)** Griffin-Lim vocoding caps quality for every model (ground truth + Griffin-Lim reaches only 3.01 UTMOS); and **(4)** models are shrunk and time-boxed to run on CPU.

The takeaway matches the paper's own conclusion: **200 sentences is the binding constraint, not the model.** Natural next steps are more training data, a neural vocoder such as HiFi-GAN in place of Griffin-Lim, and full-size models trained on GPU.